

Hackathon Project Phases Template that ensures students can complete it efficiently while covering all six phases. The template is structured to capture essential information without being time-consuming.

Hackathon Project Phases Template

Project Title: Logocraft

Team Name: Tech-Talents

Team Members:

- Y.Ramya Sri
 - V.Tejasri
 - T.Sowmya
 - Y.Hemanya
 - T.Meenakshi
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Phase-1: Brainstorming & Ideation

Objective:

Develop a user-friendly platform (LogoCraft) with AI-powered Diffusion technology. Allow users to input descriptions, keywords, and color schemes for custom logos. Generate logos based on user input and provide customization options. Enable users to download and integrate their custom logos.

Key Points:

1. Problem Statement:

- In today's competitive market, establishing a memorable brand identity through a captivating logo is crucial for businesses.
- However, many companies face challenges in creating unique and compelling logos that accurately represent their brand values and vision.

2. Proposed Solution:

- LogoCraft is an AI-powered logo design platform that leverages Generative AI (GenAI) technology to generate custom logos for businesses.

3. Target Users:

- Small Businesses: Sole proprietors, startups, and small businesses looking for

affordable logo design solutions.

- **Entrepreneurs:** Individuals launching new businesses or side hustles, requiring unique logo designs.
- **Marketing Agencies:** Agencies seeking to offer custom logo design services to their clients.

4.Expected Outcome:

- **Established Brand Identity:** LogoCraft establishes itself as a reputable brand in the logo design industry.
 - **Strategic Partnerships:** LogoCraft forms partnerships with design agencies, marketing firms, and other relevant businesses.
 - **Continuous Improvement:** The platform undergoes regular updates, incorporating user feedback and emerging design trends.
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Phase-2: Requirement Analysis

Objective:

- Develop a functional GenAI model that generates unique logo designs based on user input.

Key Points:

1. Technical Requirements:

- Programming Language: Python
- Backend: Google Gemini Flash API
- Frontend: Streamlit Web Framework
- Database: Not required initially (API-based queries)

2. Functional Requirements:

User Management

- User registration and login
- User profile management

Logo Design

- Logo design input and generation
- Logo design refinement

Design Customization

- Color palette selection
- Typography selection
- Logo icon selection

Design Download and Sharing

- Logo design download
- Logo design sharing

Payment and Subscription

- Payment gateway integration
- Subscription management

Admin Dashboard

- User management
- Design management
- Subscription management

Security and Backup

- Data encryption
- Regular backups

3. Constraints & Challenges:

- Integrating GenAI technology can be complex and require significant expertise.

- The platform must be able to handle a large volume of users and logo design requests.
 - Storing and managing large amounts of user data and logo designs can be challenging.
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Phase-3: Project Design

Objective:

- Create the architecture and user flow

Key Points:

1. System Architecture Diagram:

Frontend

- Built using React, Redux, and Material-UI
- Responsible for user interface and user experience
- Communicates with backend API using RESTful API calls

Backend

- Built using Node.js, Express.js, and MongoDB
- Responsible for handling user requests, storing data, and integrating with GenAI model
- Uses RESTful API calls to communicate with frontend

GenAI Model

- Built using TensorFlow and PyTorch
- Responsible for generating logo designs based on user input
- Trained on a dataset of logos and design elements

2. User Flow:

• User Registration

1. User visits LogoCraft website and clicks "Sign Up"
2. User enters email, password, and other details
3. System sends verification email to user
4. User verifies email address
5. System creates user account and logs user in

• User Login

1. User visits LogoCraft website and clicks "Log In"

2. User enters email and password
3. System authenticates user and logs user in

- **Logo Design Download**

1. User clicks "Download Logo" button
2. System generates final logo design file
3. User downloads final logo design file

3. **UI/UX Considerations:**

1. Logo Design Canvas: A central canvas for displaying and editing logo designs.
 2. Design Tools: A set of intuitive tools for customizing logo designs (e.g., color picker, typography selector).
 3. Navigation Menu: A clear and concise menu for accessing platform features.
 4. Feedback Mechanisms: Visual feedback for user interactions
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Phase-4: Project Planning (Agile Methodologies)

Objective:

- Break down the tasks using Agile methodologies.

Key Points:

1. **Sprint Planning:**

Sprint 1 – Setup & Integration (Day 1)

(☐ High Priority) Set up the environment & install dependencies.

(☐ High Priority) Integrate Google Gemini API.

(☐ Medium Priority) Build a basic UI with input fields.

Sprint 2 – Core Features & Debugging (Day 2)

(☐ High Priority) Implement search & comparison functionalities. (☐ High Priority) Debug API issues & handle errors in queries.

Sprint 3 – Testing, Enhancements & Submission (Day 2)

(☐ Medium Priority) Test API responses, refine UI, & fix UI bugs.

(☐ Low Priority) Final demo preparation & deployment.

Phase-5: Project Development

Objective:

- Code the project and integrate components.

Key Points:

1. Technology Stack Used:

- Frontend: Streamlit
- Backend: Google Gemini Flash API
- Programming Language: Python

2. Development Process:

- Implement API key authentication and Gemini API integration.
- Develop vehicle comparison and maintenance tips logic.
- Optimize search queries for performance and relevance.

3. Challenges & Fixes:

- Challenge: Delayed API response times. Fix: Implement caching to store frequently queried results
 - Challenge: Limited API calls per minute. Fix: Optimize queries to fetch only necessary data.
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Phase-6: Functional & Performance Testing

Objective:

- Ensure the project works as expected.

Key Points:

• Test Cases Executed:

1. Simulate multiple users creating and customizing logo designs concurrently
2. Measure design generation and rendering times under various loads
3. Test design download and sharing performance with different file sizes and formats
4. Evaluate payment processing and subscription management performance under heavy loads

• Final Validation:

1. Test Coverage: All test plans and test cases have been executed.
2. Functional Compliance: The product functions as expected and meets requirements.

3. Defect Management: Critical bugs have been resolved, and a plan is in place for non-critical issues.

Final Submission

1. **Project Report Based on the templates**
 2. **Demo Video (3-5 Minutes)**
 3. **GitHub/Code Repository Link**
 4. **Presentation**
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